

1D Kinematics



One-Dimensional Motion



Helpful Videos

- One-Dimensional Kinematics – Physics Concept Trailer:
<https://www.youtube.com/watch?v=bkbG8BJsInE>
- Position/Velocity/Acceleration Part 1: **Definitions**
<https://www.youtube.com/watch?v=4dCrkp8qgLU>
- Position/Velocity/Acceleration Part 2: **Graphical Analysis**
<https://www.youtube.com/watch?v=bqf8m7xNvLg>
- Professor Dave – Kinematics Part 1: Horizontal Motion
<https://www.youtube.com/watch?v=hpWuZh6oTew>
- Professor Dave – Kinematics Part 2: Vertical Motion
https://www.youtube.com/watch?v=K8ilu2OW7_E
- Bozeman Science – Position, Velocity and Acceleration
<https://www.youtube.com/watch?v=mXbKlYSEvF8>

Learning objectives

- Define common measurements associated with moving objects
- Relate position, velocity, and acceleration
- Determine position/velocity/acceleration using other given information
- Interpret graphs of position vs time and velocity vs time



Chapter 3: Outline

3.1 Position, Displacement, and Average Velocity

3.2 Instantaneous Velocity and Speed

3.3 Average and Instantaneous Acceleration

**Concepts &
Definitions**

3.4 Motion with Constant Acceleration

3.5 Free Fall Motion

**Physics Laws
& Formulas**

Chapter 3: Concepts & Definitions

Displacement

$$\Delta x = x_f - x_i$$

Total displacement

$$\Delta x_{\text{Total}} = \sum \Delta x_i$$

Average velocity (for constant acceleration)

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Instantaneous velocity

$$v(t) = \frac{dx(t)}{dt}$$

Average speed

$$\text{Average speed} = \bar{s} = \frac{\text{Total distance}}{\text{Elapsed time}}$$

Instantaneous speed

$$\text{Instantaneous speed} = |v(t)|$$

Average acceleration

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}$$

Instantaneous acceleration

$$a(t) = \frac{dv(t)}{dt}$$

Chapter 3: Physics Laws & Formulas

▪ 1D Motion Kinematics:

Position from average velocity

$$x = x_0 + \bar{v}t$$

Average velocity

$$\bar{v} = \frac{v_0 + v}{2}$$

Velocity from acceleration

$$v = v_0 + at \quad (\text{constant } a)$$

Position from velocity and acceleration

$$x = x_0 + v_0t + \frac{1}{2}at^2 \quad (\text{constant } a)$$

Velocity from distance

$$v^2 = v_0^2 + 2a(x - x_0) \quad (\text{constant } a)$$

▪ Free-Fall Motion: $a_y = -g = -9.8 \text{ m/s}^2$ (downward):

Velocity of free fall

$$v = v_0 - gt \quad (\text{positive upward})$$

Height of free fall

$$y = y_0 + v_0t - \frac{1}{2}gt^2$$

Velocity of free fall from height

$$v^2 = v_0^2 - 2g(y - y_0)$$

Chapter 3: Physics Laws & Formulas

Graphical Analysis:

Acceleration–time ($a - t$) graph

Velocity–time ($v - t$) graph

Position–time ($x - t$) graph

Velocity from acceleration

$$v(t) = \int a(t)dt + C_1$$

Position from velocity

$$x(t) = \int v(t)dt + C_2$$

Position

Position x : location of particle with respect to chosen reference point

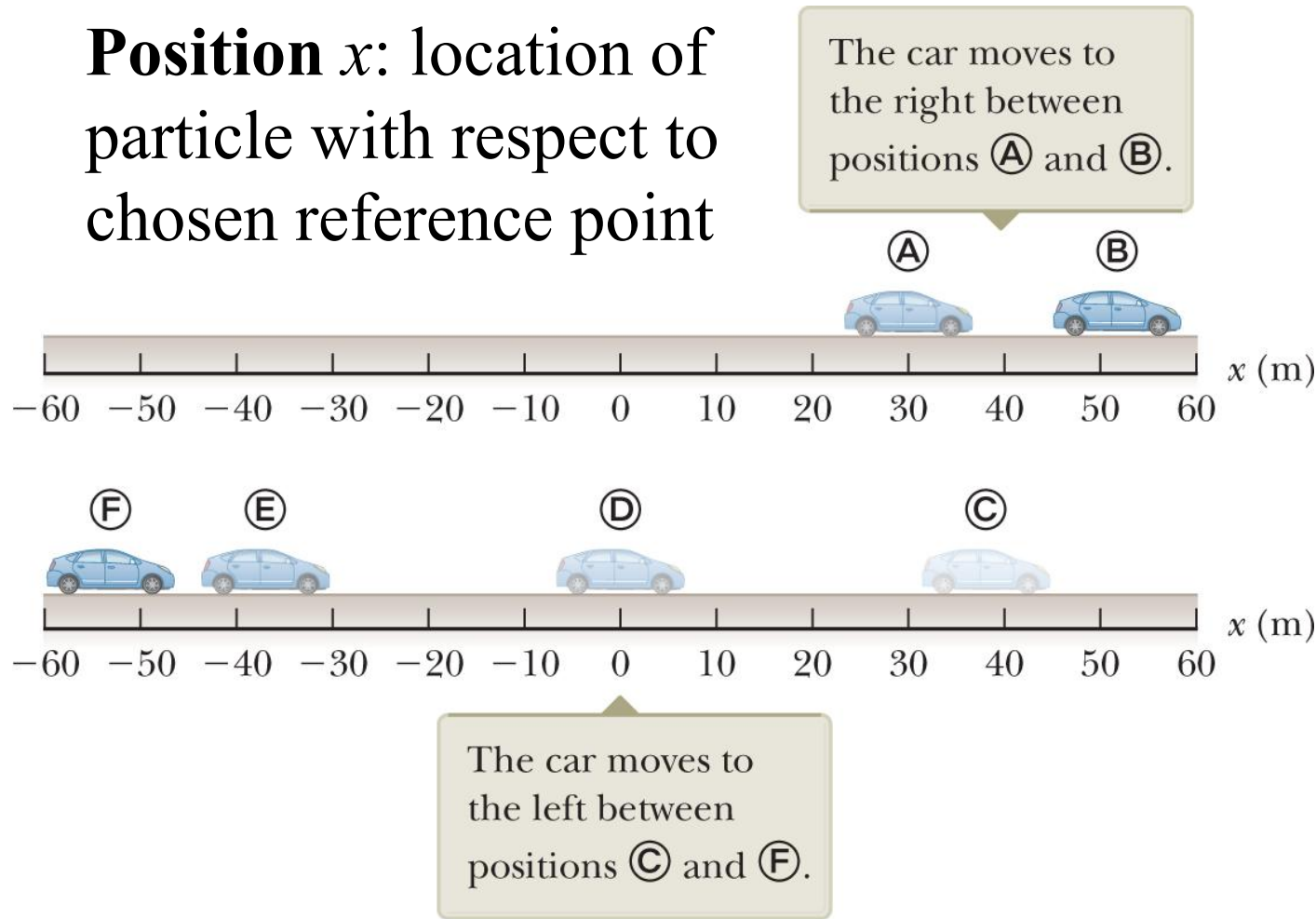
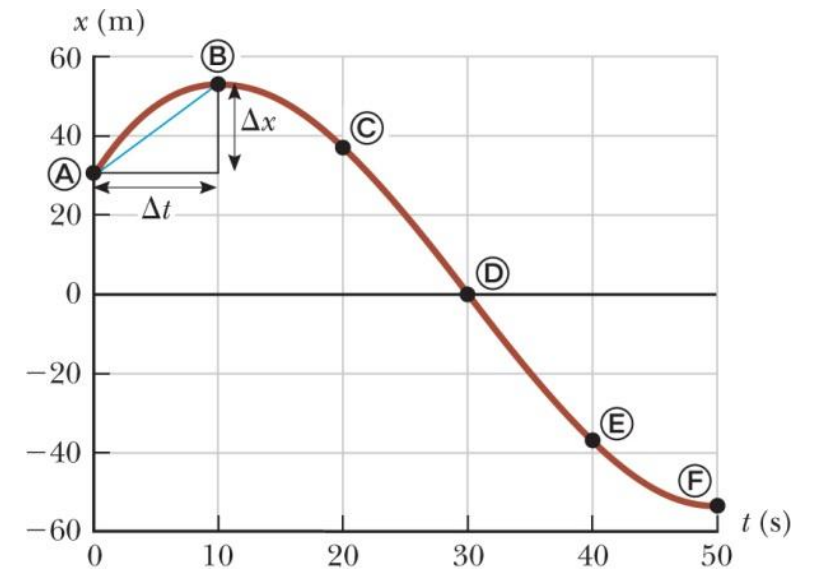


TABLE 2.1 Position of the Car at Various Times

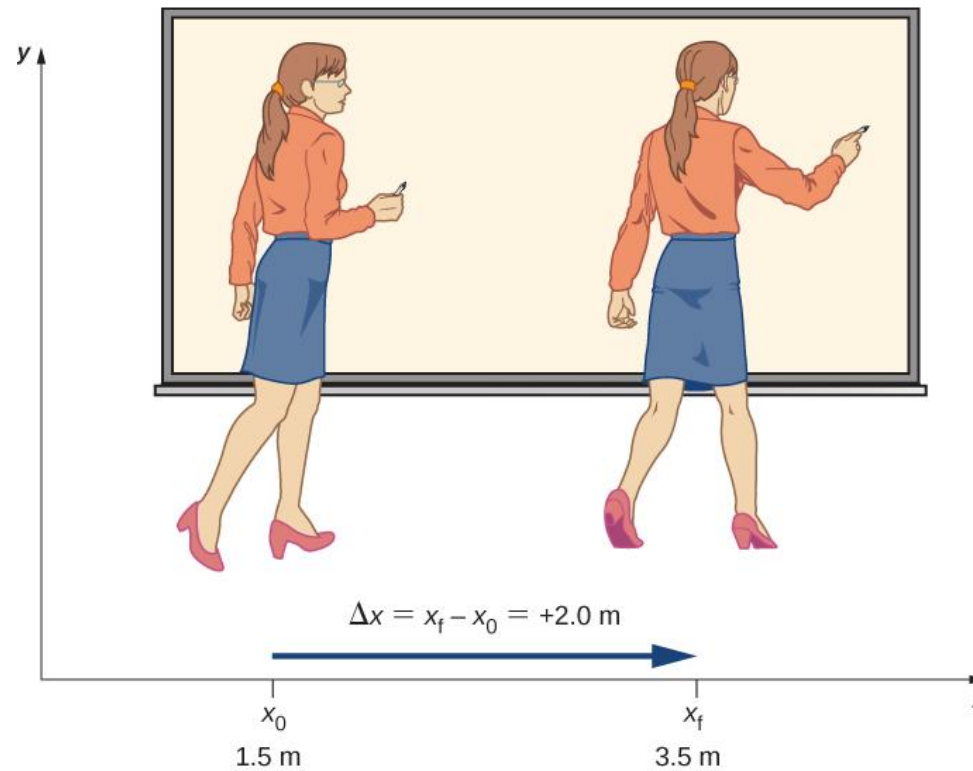
Position	t (s)	x (m)
(A)	0	30
(B)	10	52
(C)	20	38
(D)	30	0
(E)	40	-37
(F)	50	-53



a

b

Displacement



Displacement

Displacement Δx is the change in position of an object:

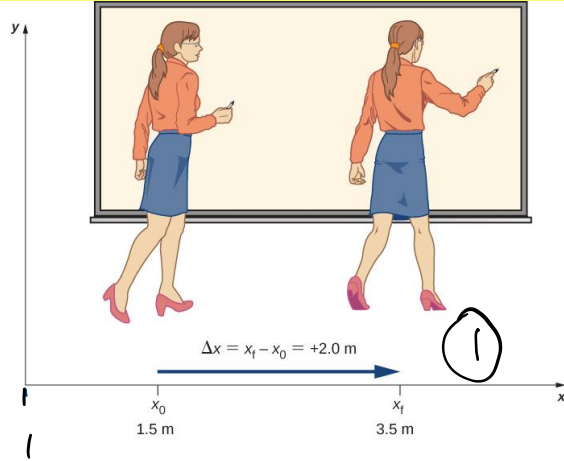
$$\Delta x = x_f - x_0,$$

where Δx is displacement, x_f is the final position, and x_0 is the initial position.

Vector quantity
(direction indicated by sign)

3.1

Displacement vs Distance Travelled



- We define **total displacement** as the sum of the individual displacements, and express this with the equation:

$$\Delta x_{\text{Total}} = \sum \Delta x_i,$$

3.2

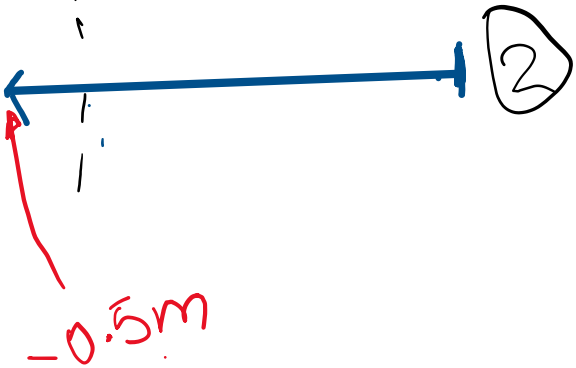
Example: $\Delta x_{\text{Total}} = \Delta x_1 + \Delta x_2 = 2 - 4 = -2 \text{ m}$

Vector quantity (direction indicated by sign)

- The **distance traveled** is the sum of the **magnitudes** of the individual displacements:

Example: $x_{\text{Total}} = |\Delta x_1| + |\Delta x_2| = 2 + 4 = 6 \text{ m}$

Scalar quantity (always positive)



CQ#1

If a car goes once around a roundabout with a radius of 20.0 m, what distance does it travel?

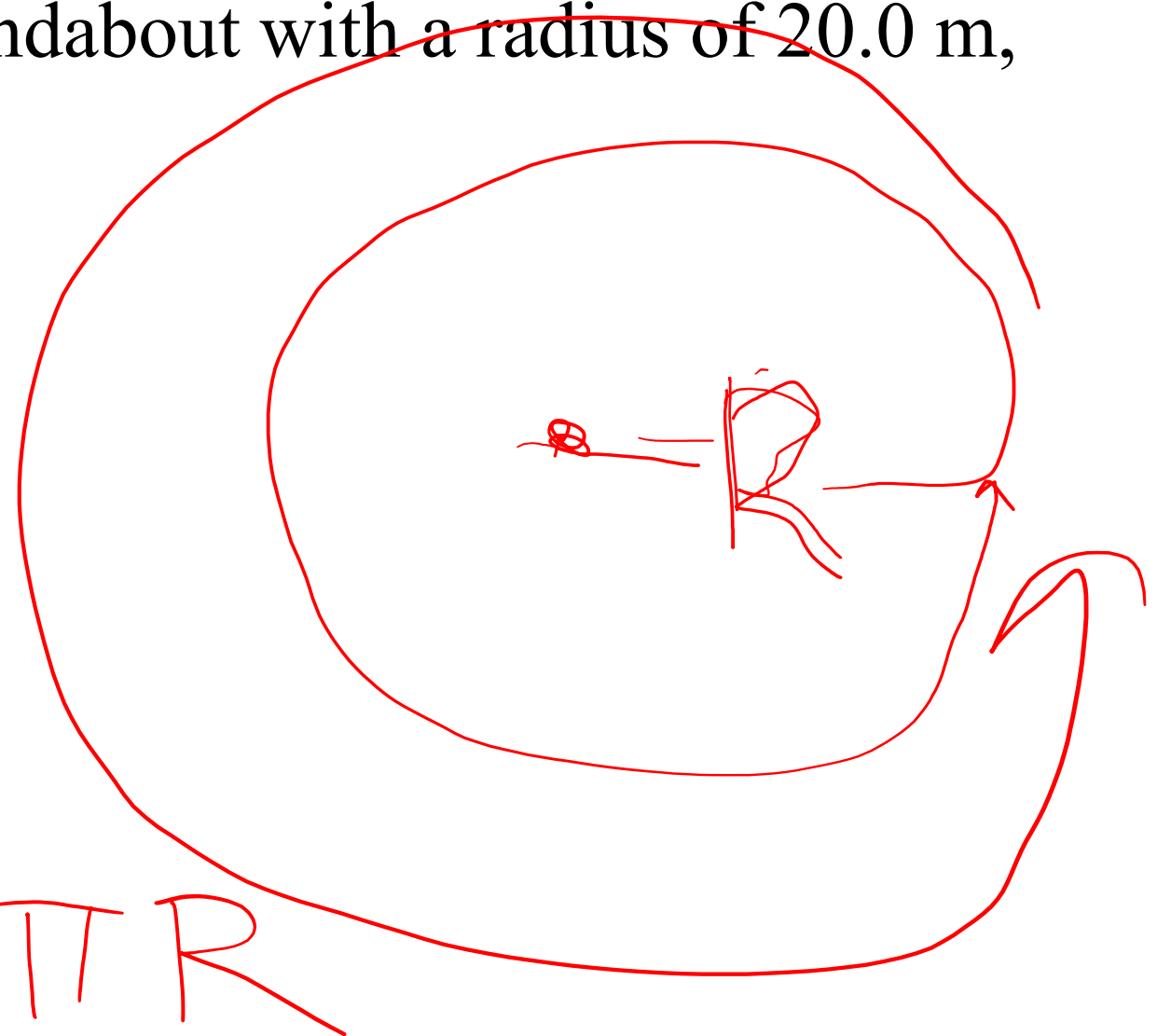
a. 20.0 m

b. 126 m

c. 0.00 m

d. 40.0 m

$$C = 2\pi R$$



CQ#2

If a car goes once around a roundabout with a radius of 20.0 m, what is the displacement?

a. 20.0 m

b. 126 m

c. 0.00 m

d. 40.0 m



ABCD card

A racecar, traveling at constant speed, completes one lap around a circular track of radius r in time t . When the car has traveled halfway around a circular track, what is the magnitude of its displacement from the starting point?

a) r [m]

b) $2r$ [m]

c) πr [m]

d) Zero [m]



Problem solving(#27 textbook)

The position of a particle moving along the x -axis is given in meters by

$$\mathbf{x(t) = (4.0 - 2.0t) \text{ m}}$$

- (a) At what time does the particle cross the origin?
- (b) What is the displacement of the particle between $t = 3.0 \text{ s}$ and $t = 6.0 \text{ s}$

Problem solving(#27 textbook)

The position of a particle moving along the x -axis is given in meters by $\mathbf{x(t) = (4.0 - 2.0t)}$ m

(a) At what time does the particle cross the origin? $\rightarrow$ Origin, $x=0$
 $y=0$

(b) What is the displacement of the particle between $t = 3.0$ s and $t = 6.0$ s

$$x_1 = x(t = 3.0 \text{ s})$$

$$x_2 = x(t = 6.0 \text{ s})$$

$$\Delta x = x_2 - x_1$$

Average Velocity vs Average Speed

Average Velocity

If x_1 and x_2 are the positions of an object at times t_1 and t_2 , respectively, then

$$\text{Average velocity} = \bar{v} = \frac{\text{Displacement between two points}}{\text{Elapsed time between two points}} \quad 3.3$$

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}.$$

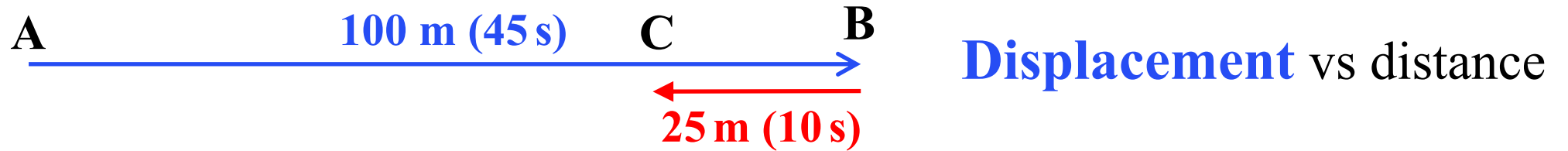
Vector quantity (sign)

- We can calculate the **average speed** by finding the total distance traveled divided by the elapsed time:

$$\text{Average speed} = \bar{s} = \frac{\text{Total distance}}{\text{Elapsed time}}. \quad 3.5$$

- **Scalar quantity (always positive)**
- Average speed is **not necessarily** the same as the magnitude of the average velocity.

Example 1: Average Velocity vs Average Speed



- Average velocity: *displacement* divided by time interval

$$\text{Average velocity: } \bar{v} = \frac{75 \text{ m}}{55.0 \text{ s}} = +1.36 \text{ m/s}$$

- Average speed: *distance* divided by time interval

$$\text{average speed} = \frac{125 \text{ m}}{55.0 \text{ s}} = 2.27 \text{ m/s}$$

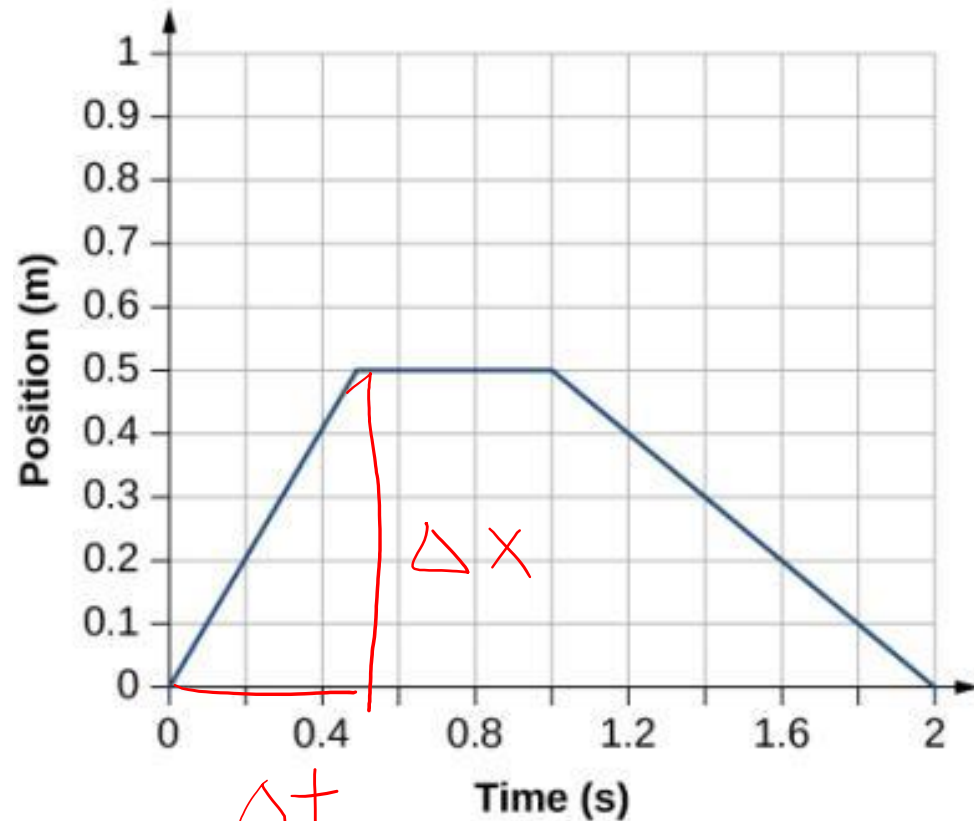
Example 2: Boat problem

During a one-hour trip, a small boat travels 80.0 km north and then travels 60.0 km east.

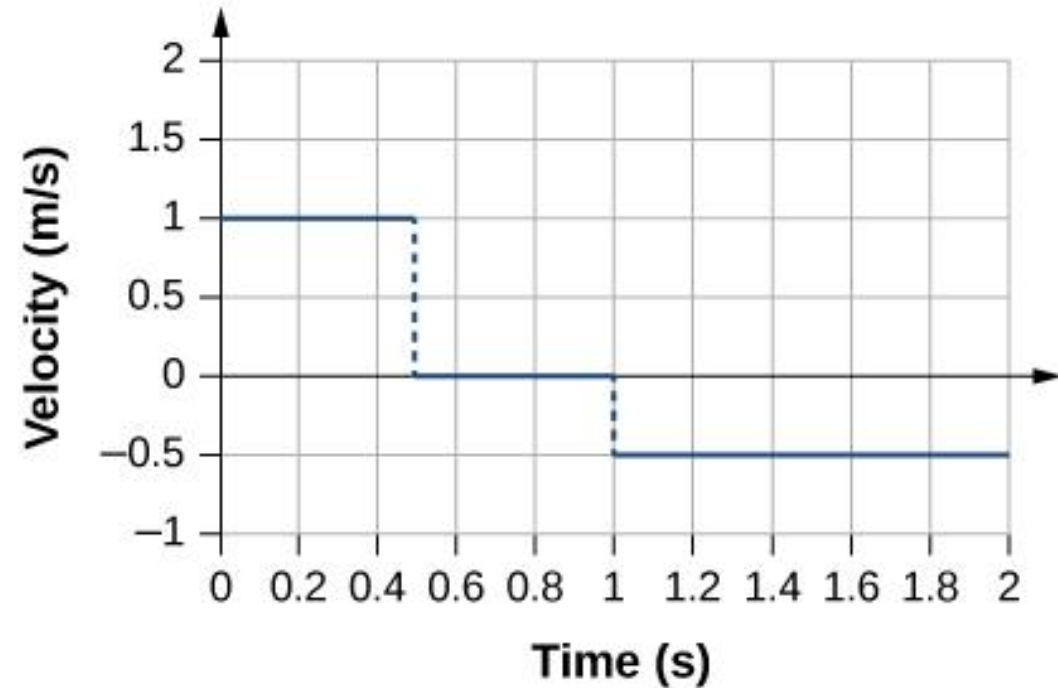
1. What is the boat's displacement for the one-hour trip? [$1.00 \times 10^5 \text{m}$]
2. What is the boat's average speed during the one-hour trip? [39.0 m/s]
3. What is the magnitude of the boat's average velocity for the one-hour trip? [28.0 m/s]
4. What is the direction of the boat's average velocity for the one-hour trip? [$53.1^\circ \text{ North of East}$]

Graphical Analysis (**constant** motion)

Position vs. Time



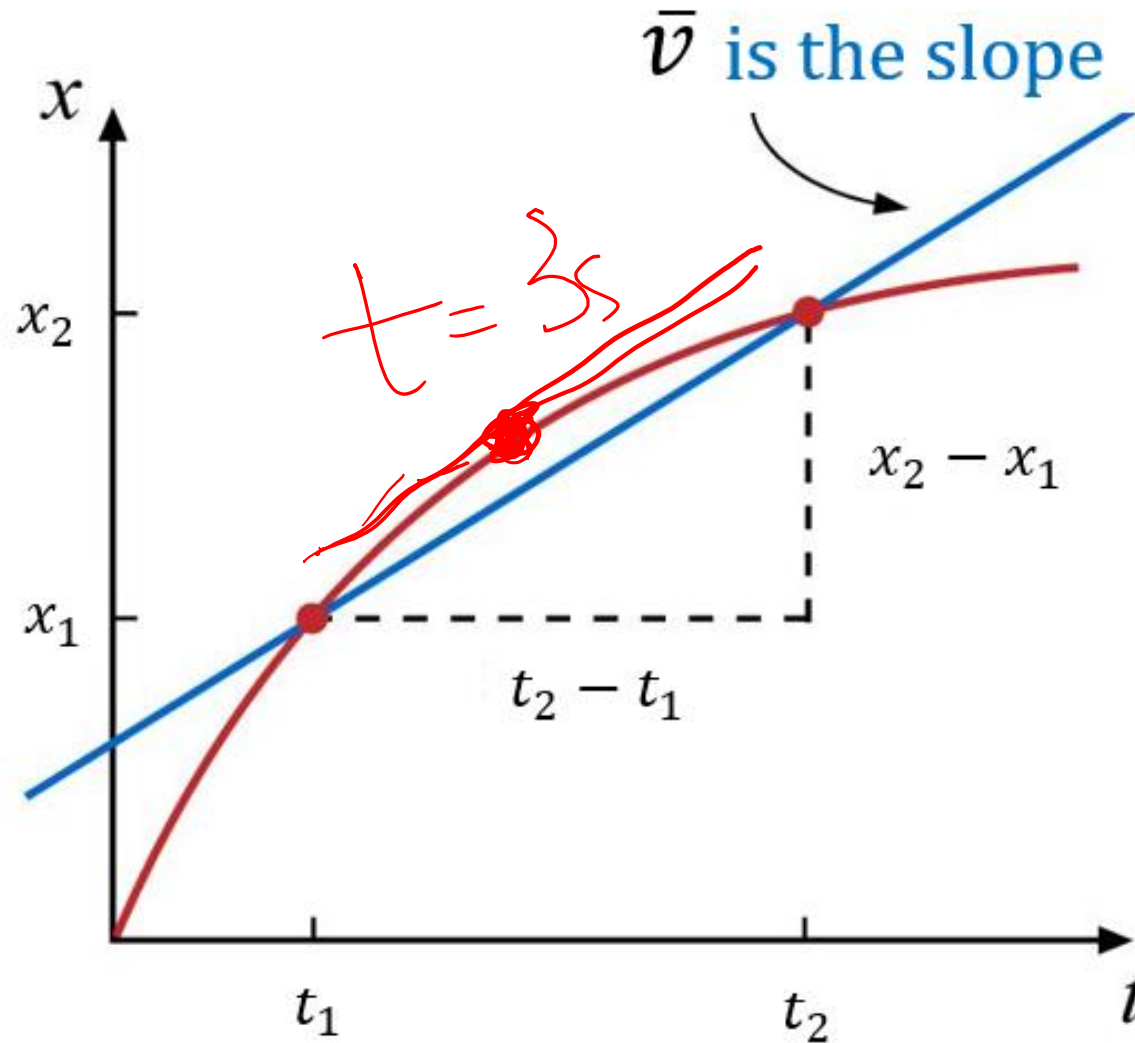
Velocity vs. Time



$$V = \frac{\Delta x}{\Delta t}$$

Slope of tangent line in the position-time graph $\rightarrow$ velocity

Graphical Analysis (**accelerated** motion)



Instantaneous Velocity

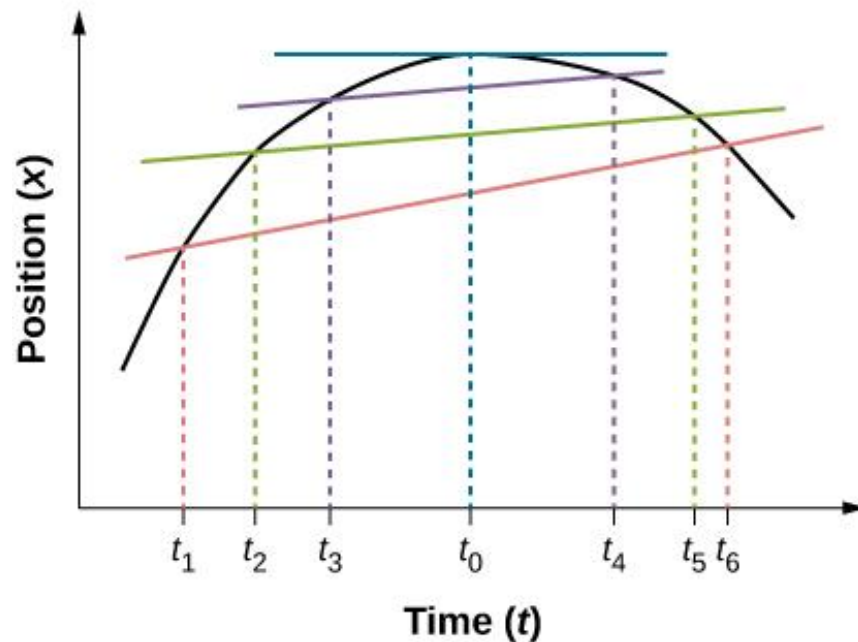
Instantaneous Velocity

The instantaneous velocity of an object is the limit of the average velocity as the elapsed time approaches zero, or the derivative of x with respect to t :

$$v(t) = \frac{d}{dt}x(t).$$

3.4

$v(t_0) = \text{slope of tangent line}$



- In a graph of **position versus time**, the instantaneous velocity is **the slope of the tangent line at a given point**. The average velocities $\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$ between times $\Delta t = t_6 - t_1$, $\Delta t = t_5 - t_2$, and $\Delta t = t_4 - t_3$ are shown. When $\Delta t \rightarrow 0$, the average velocity approaches the instantaneous velocity at $t = t_0$.

Instantaneous **Velocity** and Instantaneous Speed

Instantaneous Velocity

The instantaneous velocity of an object is the limit of the average velocity as the elapsed time approaches zero, or the derivative of x with respect to t :

$$v(t) = \frac{d}{dt}x(t). \quad 3.4$$

Vector quantity (sign)

- We can calculate the **instantaneous speed** from the magnitude of the instantaneous velocity:

$$\text{Instantaneous speed} = |v(t)|. \quad 3.6$$

Scalar quantity (always positive)

Problem solving (#35 textbook)

A particle moves along the x -axis in meters according to $\mathbf{x(t) = (10t - 2t^2) \text{ m}}$

- (a) What is the instantaneous velocity at $t = 2 \text{ s}$ and $t = 4 \text{ s}$?
- (b) What is the instantaneous speed at these times?
- (c) What is the average velocity between 2 s and 4 s ?

Problem solving (#35 textbook)

(a) What is the instantaneous velocity at $t = 2$ s and $t = 4$ s? $x(t) = (10 t - 2 t^2)$ m

$$v(t) = \frac{dx(t)}{dt}$$

Problem solving (#35 textbook)

(a) What is the instantaneous velocity at $t = 2$ s and $t = 4$ s?

$$\begin{aligned}v(t) &= \frac{dx(t)}{dt} \\&= \frac{d}{dt}(10t - 2t^2) \\&= (10 - 4t)\end{aligned}$$

At time $t = 2$ s,

$$\begin{aligned}v(t = 2\text{ s}) &= [10 - 4(2)] \\&= [10 - 8] \\&= 2 \text{ m/s}\end{aligned}$$

At time $t = 4$ s,

$$\begin{aligned}v(t = 4\text{ s}) &= [10 - 4(4)] \\&= [10 - 16] \\&= -6 \text{ m/s}\end{aligned}$$

Problem solving (#35 textbook)

(b) What is the instantaneous speed at these times?

At time $t = 2s$,

$$|v(t = 2s)| = 2m/s$$

At time $t = 4s$,

$$|v(t = 4s)| = 6 \text{ m/s}$$

Problem solving (#35 textbook)

$$x(t) = (10 t - 2 t^2) \text{ m}$$

(c) What is the average velocity between 2 s and 4 s?

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \dots (1)$$

Problem solving (#35 textbook)

$$x(t) = (10 t - 2 t^2) \text{ m}$$

(c) What is the average velocity between 2 s and 4 s?

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \dots (1)$$

$$x_2: x(t = 4s) = [10 (4) - 2 (4)^2] \text{ m} = 8 \text{ m}$$

$$x_1: x(t = 2s) = [10 (2) - 2 (2)^2] \text{ m} = 12 \text{ m}$$

$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{8 - 12}{4 - 2} = \frac{-4}{2}$$

$$\bar{v} = -2 \text{ m/s}$$

Graphical Analysis Instantaneous velocity (Example 3.4)

Consider the motion of a particle for $t > 0$ in which position is $\mathbf{x(t) = (3.0\,t - 3t^2)\,m}$

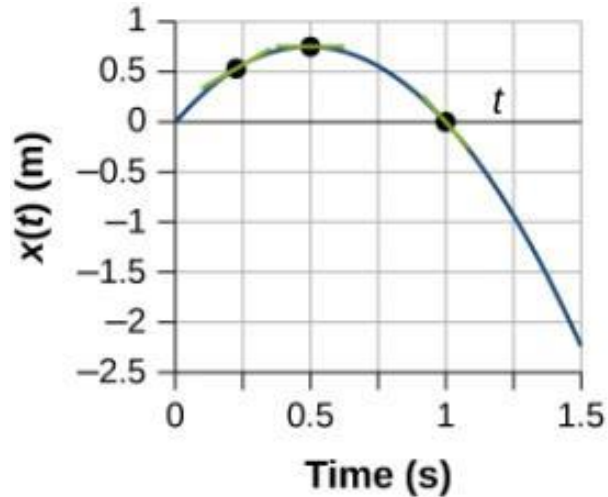
a) What is the **instantaneous velocity at** $t = 0.25\,\text{s}$, $t = 0.50\,\text{s}$ and $t = 1.0\,\text{s}$?

b) What is the **speed** of the particle at these times?

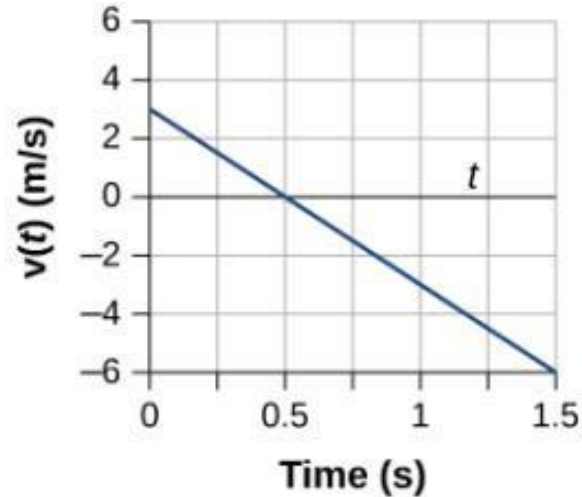
	$x(\text{m})$	$v(\text{m/s})$
$t = 0.25\,\text{s}$	0.56	1.5
$t = 0.50\,\text{s}$	0.75	0.0
$t = 1.00\,\text{s}$	0.00	-3.0

Graphical Analysis Instantaneous velocity (Example 3.4)

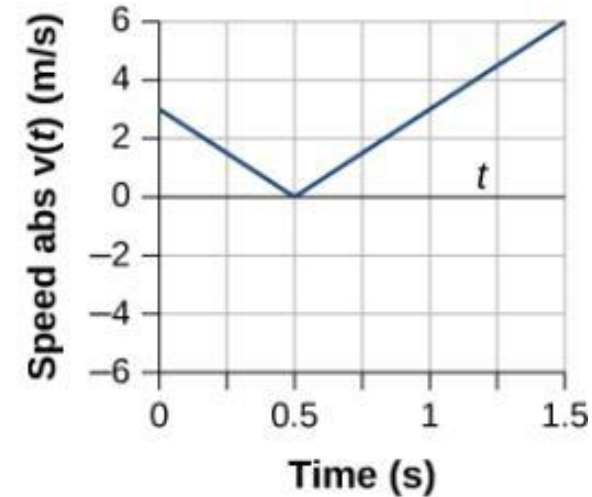
Consider the motion of a particle for $t > 0$ in which position is $\mathbf{x(t) = (3.0\,t - 3t^2)\,m}$



(a) Position



(b) Velocity



(c) Speed

(a) Position: $x(t)$ versus time.

(b) Velocity: $v(t)$ versus time. The slope of the position graph is the velocity. A rough comparison of the slopes of the tangent lines in (a) at 0.25 s, 0.50 s, and 1.0 s with the values for velocity at the corresponding times indicates they are the same values.

(c) Speed: $|v(t)|$ versus time. Speed is always a positive number.

Average and Instantaneous Acceleration

Average Acceleration

Average acceleration is the rate at which velocity changes:

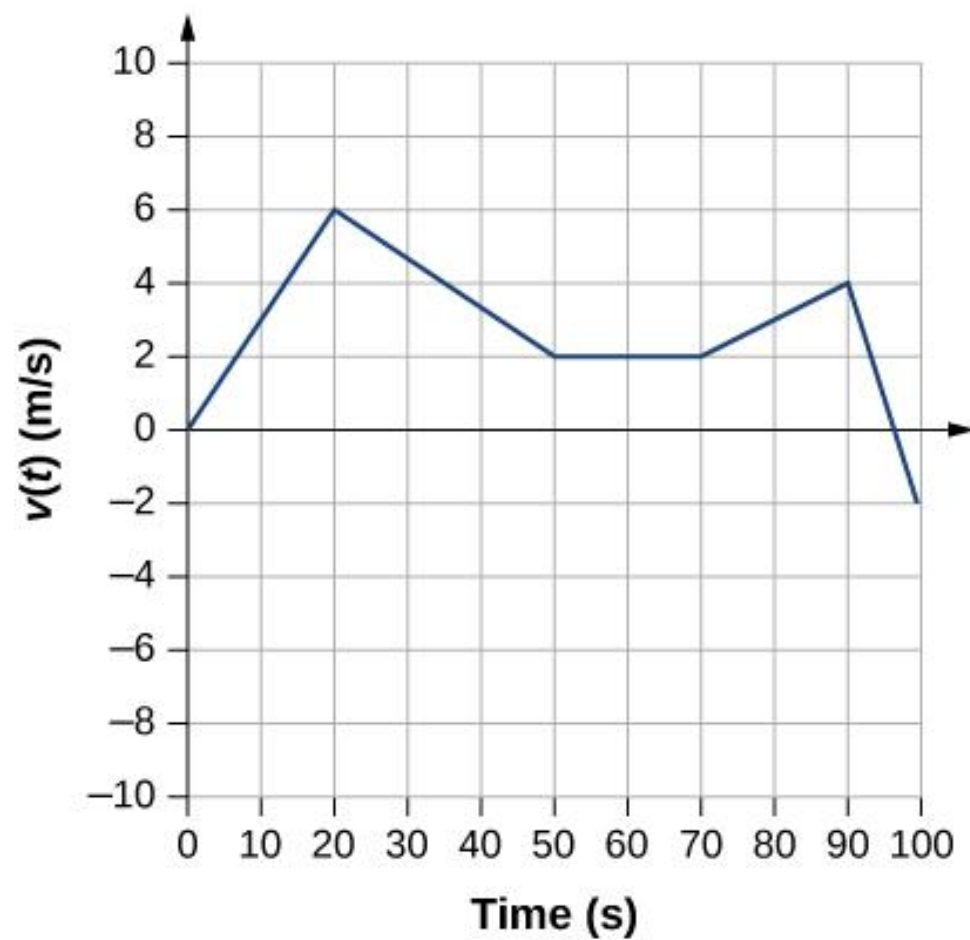
$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}, \quad 3.8$$

where $\bar{a}$ is **average acceleration**, v is velocity, and t is time. (The bar over the a means *average* acceleration.)

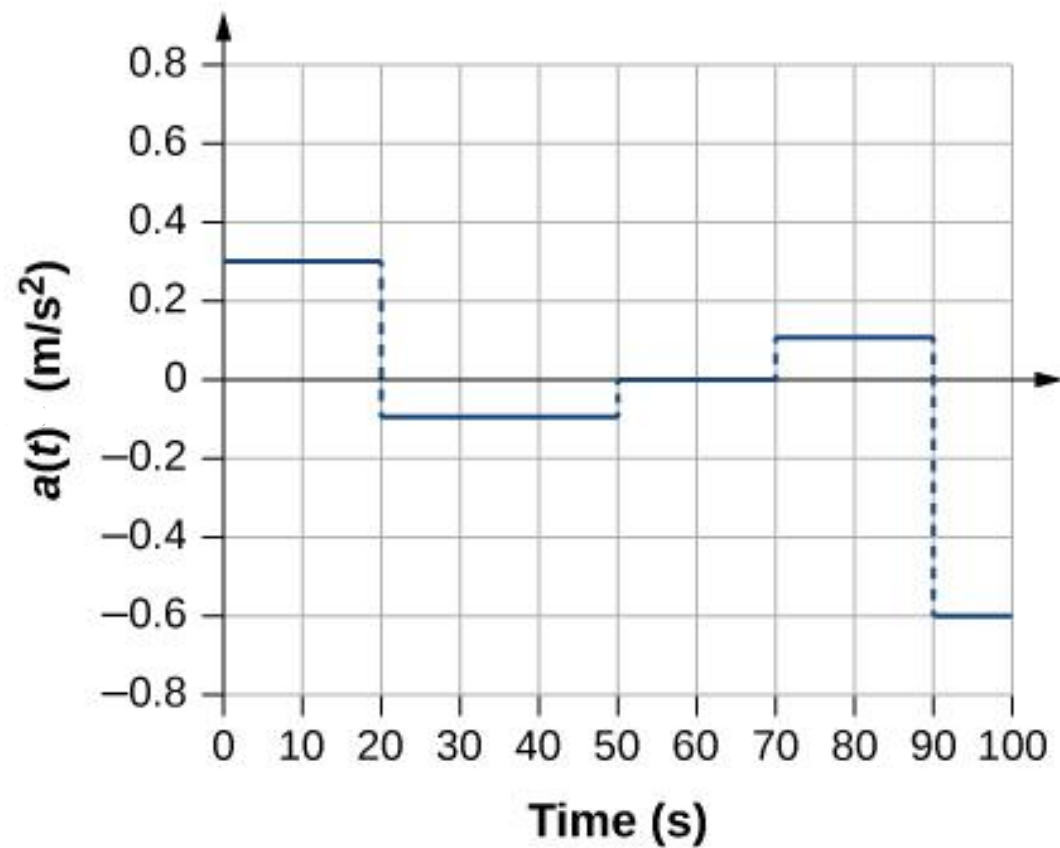
Acceleration as a Vector

Acceleration is a vector in the same direction as the *change* in velocity, Δv . Since velocity is a vector, it can change in magnitude or in direction, or both. Acceleration is, therefore, a change in speed or direction, or both.

Velocity vs. Time



Acceleration vs. Time

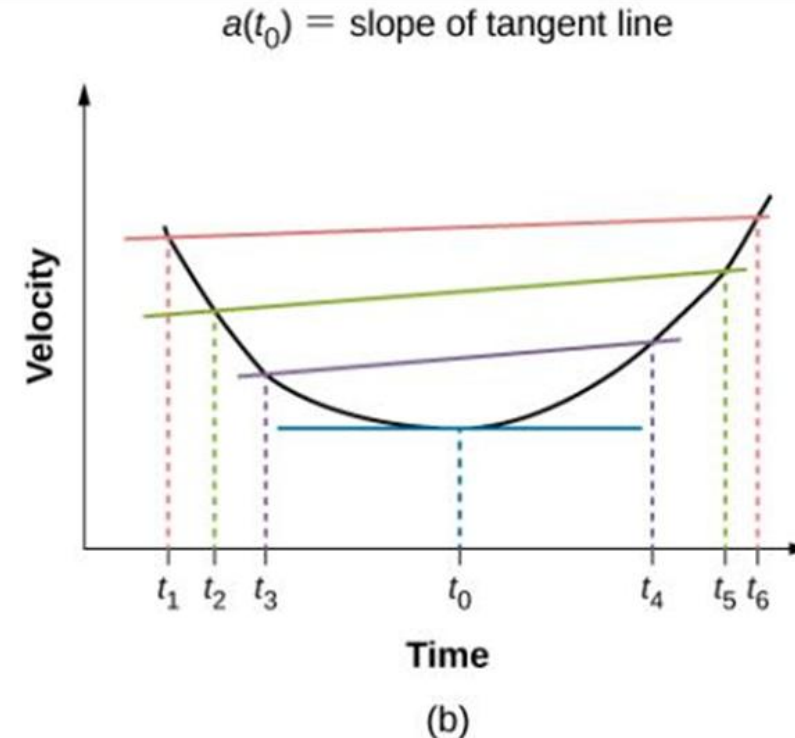
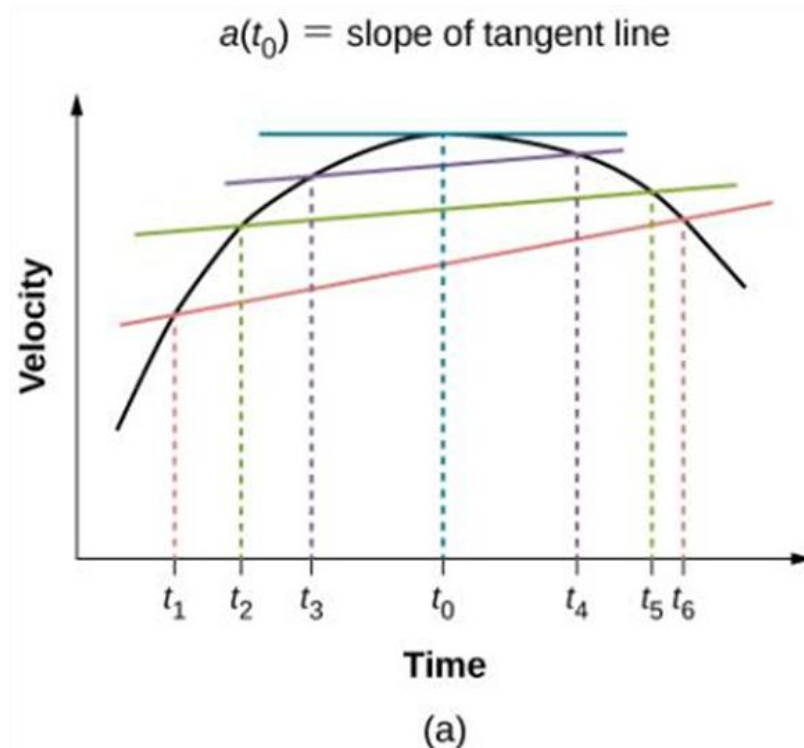


Average and Instantaneous Acceleration

Instantaneous acceleration a , or *acceleration at a specific instant in time*, is obtained using the same process discussed for instantaneous velocity. That is, we calculate the average acceleration between two points in time separated by Δt and let Δt approach zero. The result is the derivative of the velocity function $v(t)$, which is **instantaneous acceleration** and is expressed mathematically as

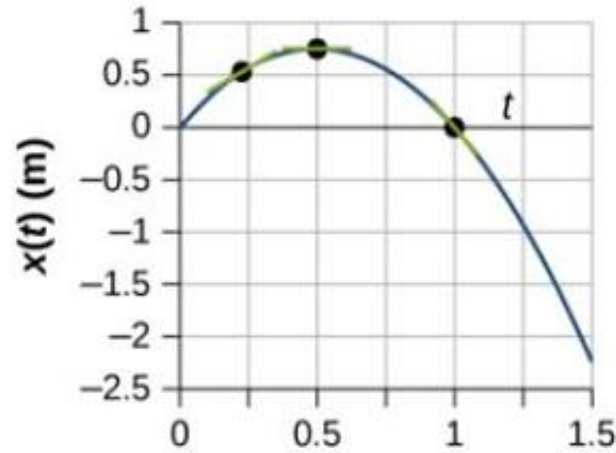
$$a(t) = \frac{d}{dt}v(t).$$

3.9

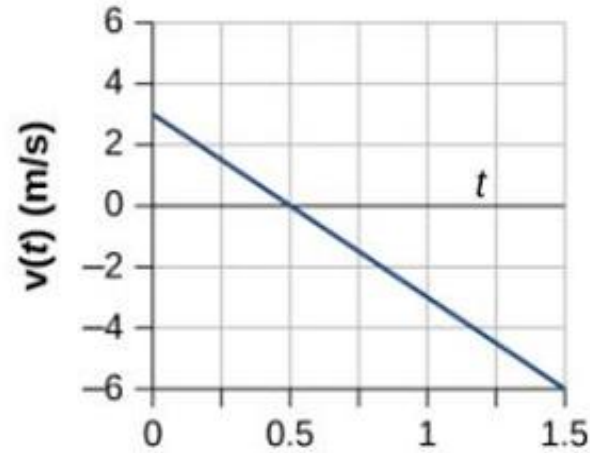


Acceleration-Time graph (Example 3.4)

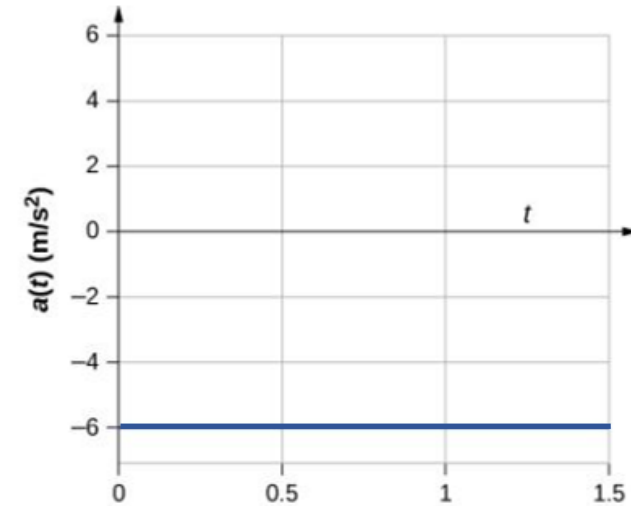
Consider the motion of a particle for $t > 0$ in which position is $x(t) = (3.0 t - 3t^2) \text{ m}$



Time (s)
(a) Position



Time (s)
(b) Velocity

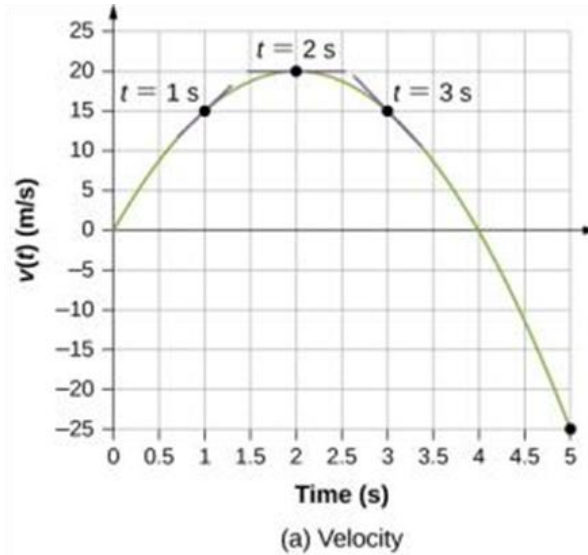


Time (s)
(c) Acceleration

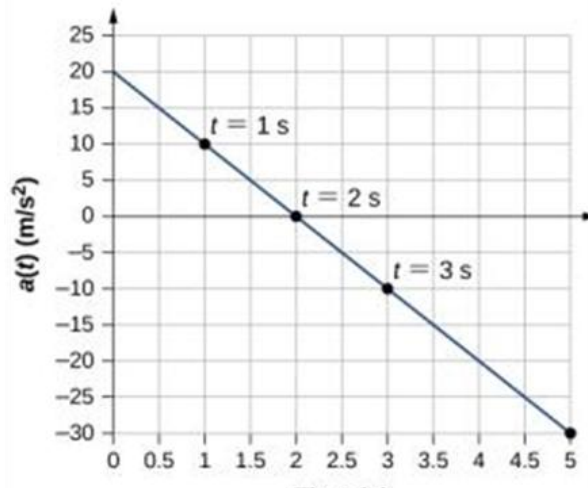
- **Slope** of the tangent line in the velocity-time graph $\rightarrow$ acceleration
- The velocity-time graph is linear and has a negative constant slope (b) that is equal to the acceleration, as shown in (c).

Acceleration-Time graph (Example 3.6)

The functional form of velocity for $t > 0$, $\mathbf{v(t) = 20\,t - 5t^2\,m/s}$



(a) Velocity-time graph. Tangent lines are indicated at times 1, 2, and 3 s. The slopes of the tangent lines are the accelerations. At $t = 3$ s, velocity is positive. At $t = 5$ s, velocity is negative, indicating the particle has reversed direction.

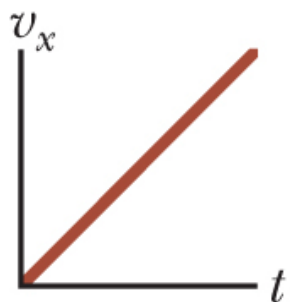


(b) Acceleration-time graph. Comparing the values of accelerations given by the black dots with the corresponding slopes of the tangent lines (slopes of lines through black dots) in (a), we see they are identical.

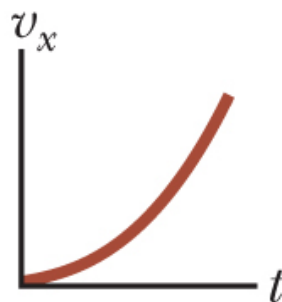
Q. Can you find position if velocity function is given $\mathbf{v(t) = 20\,t - 5t^2\,m/s}$???

Quick Quiz?

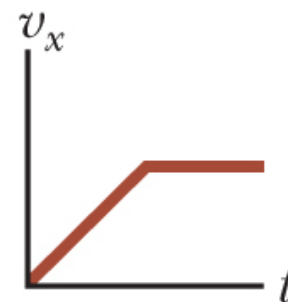
In the figure, match each v_x – t graph on the top with the a_x – t graph on the bottom that best describes the motion.



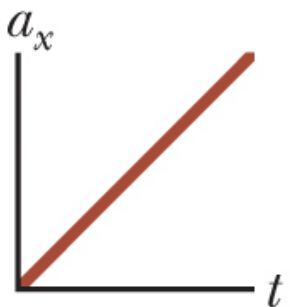
a



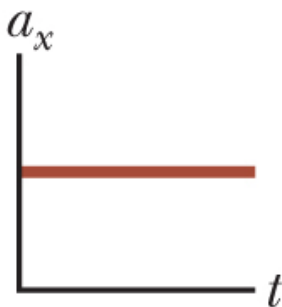
b



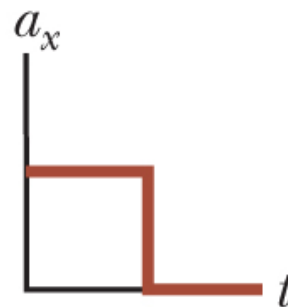
c



d



e



f

(a)–(e), (b)–(d), (c)–(f)

Kinematic Equation (constant “ a ”)

Equations for kinematic relationships, starting from the definitions of displacement, velocity, and acceleration.

$$x = x_0 + \bar{v}t$$

$$\bar{v} = \frac{v_0 + v}{2}$$

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

Displacement and Position from Velocity

To get our first two equations, we start with the definition of average velocity:

$$\bar{v} = \frac{\Delta x}{\Delta t}.$$

Substituting the simplified notation for Δx and Δt yields

$$\bar{v} = \frac{x - x_0}{t}.$$

Solving for x gives us

$$x = x_0 + \bar{v}t, \quad 3.10$$

where the average velocity is

$$\bar{v} = \frac{v_0 + v}{2}. \quad 3.11$$

Solving for Final Velocity from Acceleration and Time

We can derive another useful equation by manipulating the definition of acceleration:

$$a = \frac{\Delta v}{\Delta t}.$$

Substituting the simplified notation for Δv and Δt gives us

$$a = \frac{v - v_0}{t} \quad (\text{constant } a).$$

Solving for v yields

$$v = v_0 + at \quad (\text{constant } a).$$

3.12

Solving for Final Position (constant a)

We can combine the previous equations to find a third equation that allows us to calculate the final position of an object experiencing constant acceleration. We start with

$$v = v_0 + at.$$

Adding v_0 to each side of this equation and dividing by 2 gives

$$\frac{v_0 + v}{2} = v_0 + \frac{1}{2}at.$$

Since $\frac{v_0 + v}{2} = \bar{v}$ for constant acceleration, we have

$$\bar{v} = v_0 + \frac{1}{2}at.$$

Now we substitute this expression for $\bar{v}$ into the equation for displacement, $x = x_0 + \bar{v}t$, yielding

$$x = x_0 + v_0t + \frac{1}{2}at^2 \quad (\text{constant } a).$$

3.13

Solving for Final Velocity from Distance and Acceleration

A fourth useful equation can be obtained from another algebraic manipulation of previous equations. If we solve $v = v_0 + at$ for t , we get

$$t = \frac{v - v_0}{a}.$$

Substituting this and $\bar{v} = \frac{v_0 + v}{2}$ into $x = x_0 + \bar{v}t$, we get

$$v^2 = v_0^2 + 2a(x - x_0) \text{ (constant } a\text{).}$$

3.14

Problem solving (Textbook) #58

Freight trains can produce only relatively small accelerations,

- a) What is the final velocity of a freight train that accelerates at a rate of 0.0500 m/s^2 for 8.00 min, starting with the initial velocity of 4.00 m/s ?

Answer: $v = 28.0 \text{ m/s}$

- b) If the train can slow down at a rate of 0.550 m/s^2 , how long will it take to come to stop from this velocity?

Answer: $t = 50.9 \text{ s}$

- c) How far will it travel in each case?

**Answer: $x_1 = 7680 \text{ m}$ and
 $x_2 = 713 \text{ m}$**

Problem solving (Textbook) #58

a) Given $a = 0.0500 \text{ m/s}^2$

$$t = 8.00 \text{ min} = (8.00 \times 60) \text{ s} = 480 \text{ s}$$

$$V_0 = 4.00 \text{ m/s}$$

$$V = ?$$

We know, $V = V_0 + at$

$$= (4.00) + (0.0500)(480)$$

$$\boxed{V = 28.0 \text{ m/s}}$$

Problem solving (Textbook) #58

$$\begin{array}{l} b) \quad V_0 = 28.0 \text{ m/s} \\ \quad \quad V = 0 \text{ m/s} \end{array} \left. \vphantom{\begin{array}{l} V_0 = 28.0 \text{ m/s} \\ V = 0 \text{ m/s} \end{array}} \right\} \text{slowing} \quad a = -0.550 \text{ m/s}^2$$

$$t = ?$$

We know, $V = V_0 + at$

$$\frac{V - V_0}{a} = \frac{\cancel{a}t}{\cancel{a}}$$

$$t = \frac{V - V_0}{a} = \frac{0 - (28.0)}{(-0.550)}$$

$$\boxed{t = 50.9 \text{ s}}$$

Problem solving (Textbook) #58

c) 1st case,

$$v_0 = 4.00 \text{ m/s}$$

$$a = 0.0500 \text{ m/s}^2$$

$$t_1 = 480 \text{ s}$$

$$x = ?$$

We know, $x = \cancel{x_0} + v_0 t + \frac{1}{2} a t^2$

$$x = (4.00)(480) + \frac{1}{2} (0.0500)(480)^2$$

$$x_1 = 7680 \text{ m}$$

0 initial position

2nd case

$$v_0 = 28.0 \text{ m/s}$$

$$a = -0.550 \text{ m/s}^2$$

$$t_2 = 50.9 \text{ s}$$

$$x = ?$$

Again, $\cancel{x_0}$

$$x = \cancel{x_0} + v_0 t + \frac{1}{2} a t^2$$

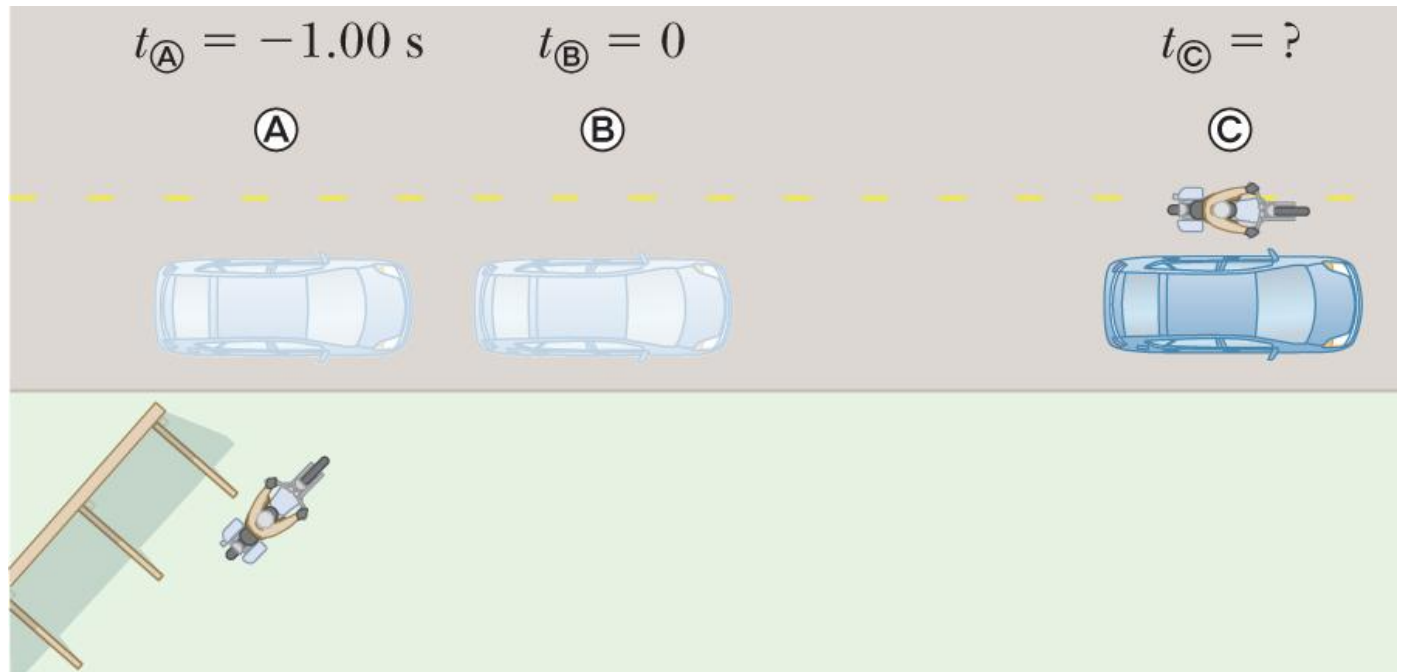
$$x = (28.0)(50.9) + \frac{1}{2} (-0.550)(50.9)^2$$

$$x_2 = 712.12 \text{ m}$$

Example 3 (Two-body Pursuit Problems)

You are driving at a constant speed of 45.0 m/s when you pass a trooper on a motorcycle hidden behind a billboard. One second after your car passes the billboard, the trooper sets out from the billboard to catch you,

accelerating at a constant rate of 3.00 m/s^2 . How long does it take the trooper to overtake your car?



Example 3 (Two-body Pursuit Problems)

$$x = x_0 + v_0 t + \frac{1}{2} a t^2 \text{ ... (3.13)}$$

For car,

$$x_{\text{car}} = x_B + v_{\text{car}} t$$

For trooper,

$$\begin{aligned} x_{\text{trooper}} &= x_0 + v_{0 \text{ (trooper)}} t + \frac{1}{2} a_{\text{trooper}} t^2 \\ &= \frac{1}{2} a t^2 \end{aligned}$$

Let 't' be the time when trooper overtake you,

$$x_{\text{trooper}} = x_{\text{car}}$$

$$\frac{1}{2} a t^2 = x_B + v_{\text{car}} t$$

$$\frac{1}{2} (3.00) t^2 = (45.0 \text{ m}) + (45.0 \text{ m/s}) t$$

Example 3 (Two-body Pursuit Problems)

$$\frac{1}{2}(3.00)t^2 = (45.0) + (45.0)t$$

$$1.50 t^2 - 45.0 t - 45.0 = 0$$

For quadratic equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$$

$$t = \frac{45.0 \pm \sqrt{(45.0)^2 - 4(1.50)(-45.0)}}{2(1.50)} = \boxed{31.0 \text{ s}}$$

Problem solving (Two-body Pursuit Problems) #96

Pablo is running in a half marathon at a velocity of 3.0 m/s . Another runner, **Jacob**, is 50 meters behind **Pablo** with the same velocity. Jacob begins to accelerate at the constant rate of 0.050 m/s^2 .

(a) How long does it take **Jacob** to catch **Pablo**?

Answer: $t = 45 \text{ s}$

(b) What is the distance covered by Jacob?

Answer: $x_J = 184.0 \text{ m}$

(c) What is the final velocity of the Jacob?

Answer: $v_J = 5.2 \text{ m/s}$



Problem solving (Two-body Pursuit Problems) #96

$$x = x_0 + v_0 t + \frac{1}{2} a t^2 \dots \textbf{(3.13)}$$

For Pablo,

$$x_P = x_{0P} + v_{0P} t = 50 + 3.0 t$$

For Jacob,

$$x_J = x_{0J} + v_{0J} t + \frac{1}{2} a_J t^2 = 0 + 3.0 t + \frac{1}{2} (0.05) t^2$$

Let 't' be the time when Jacob catches Pablo,

$$a) x_P = x_J$$

(Solve the above equation for time)

$$t = \dots s$$

Problem solving (Two-body Pursuit Problems) #96

$$\text{b) } x_J = 0 + 3.0 t + \frac{1}{2} (0.05) t^2$$

(Substitute the value of time from “ part a” in the above equation for x_J)

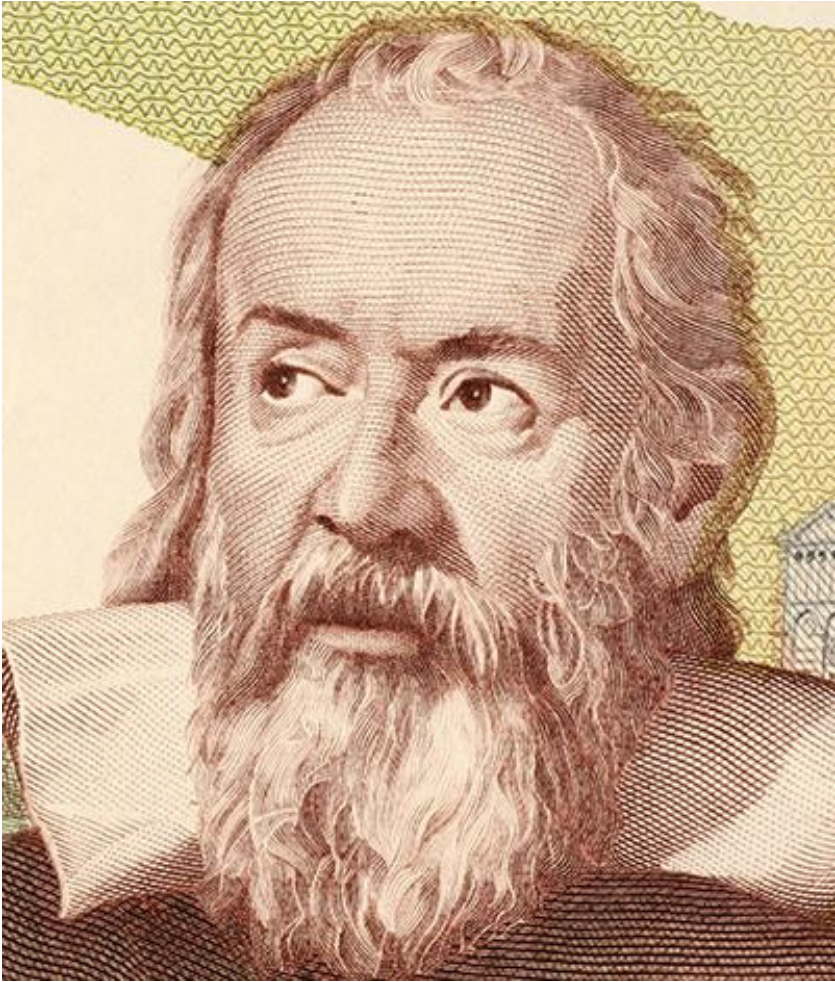
$$x_J = \cdots m$$

$$\text{c) } v_J = v_{0J} + a_J t$$

(Substitute the value of v_{0J} and time from “part a” in the above equation for v_J)

$$v_J = \cdots m/s$$

Free Fall



Galileo Galilei



Leaning Tower of Pisa

Free Fall

- A freely falling object is any object moving freely under the influence of gravity alone, **regardless of its initial velocity**.
- Objects thrown upward or downward, and those released from rest are all falling freely once they are released.
- We define the **upward direction as positive**, and then constant acceleration: $a_y = -g = -9.80 \text{ m/s}^2$, where the negative sign means that it is downward.
- Gravitational acceleration: $g = 9.80 \text{ m/s}^2$ (g is only the magnitude and always positive).

A bowling ball and a feather fall with the same constant acceleration if air resistance is negligible



Kinematic Equations for Free Fall Motion

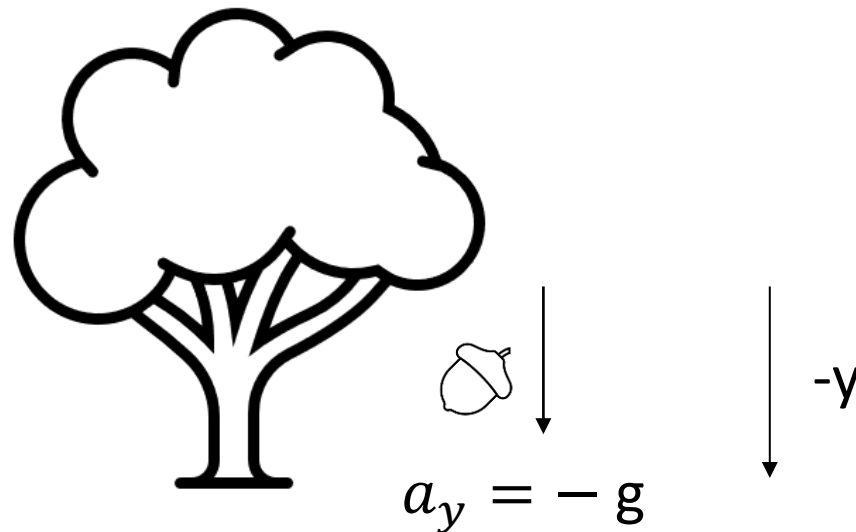
Kinematic Equations for Objects in Free Fall

We assume here that acceleration equals $-g$ (with the positive direction upward).

$$v = v_0 - gt \quad 3.15$$

$$y = y_0 + v_0 t - \frac{1}{2}gt^2 \quad 3.16$$

$$v^2 = v_0^2 - 2g(y - y_0) \quad 3.17$$



$$v = v_y$$

$$a = a_y = -g$$

$$g = 9.80 \text{ m/s}^2$$

ABCD card

A wrench and a feather are both dropped at the same time, which will hit the ground first? (assume no air resistance)

- a. Wrench
- b. Feather
- c. Same time

Free Fall

Graphical representations:

Acceleration vs time (a_y vs t) graph

Velocity vs time (v_y vs t) graph

Position vs time (y vs t) graph

Three situations:

Initial Velocity is upward, zero, or downward.

CQ#3

If I toss an object upward with a velocity of 3 m/s, what is its velocity at its highest point?

- a. 3.00 m/s
- b. -9.80 m/s^2
- c. 0.00 m/s
- d. -3.00 m/s

CQ#4

What happens to the **acceleration** of a ball after it is thrown upward into the air? (Neglect air resistance.)

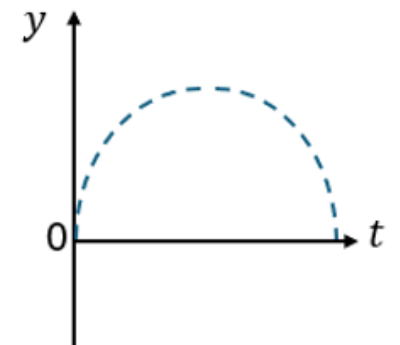
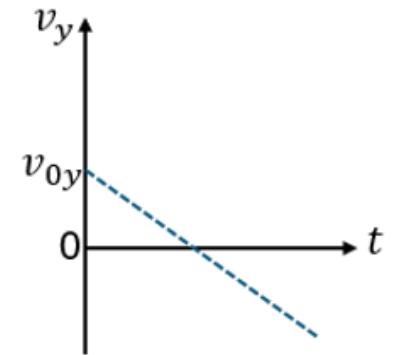
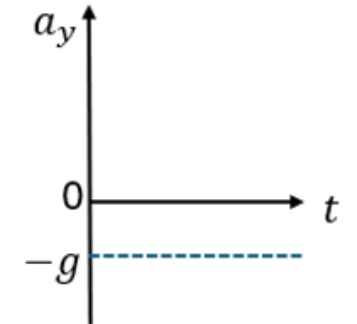
- (a) increases
- (b) decreases
- (c) increases and then decreases
- (d) decreases and then increases
- (e) remains the same

ABCD card

What happens to the speed of a ball after it is thrown upward into the air? (Neglect air resistance.)

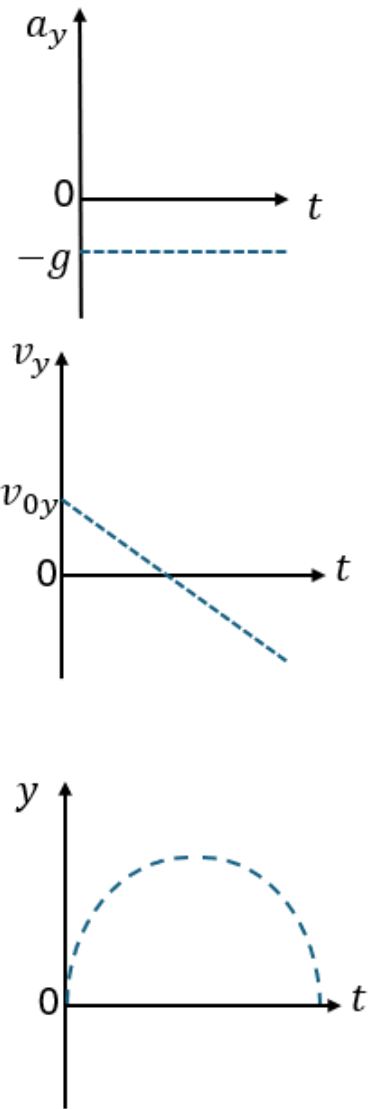
- (a) increases
- (b) decreases
- (c) increases and then decreases
- (d) decreases and then increases
- (e) remains the same

Initial velocity upward

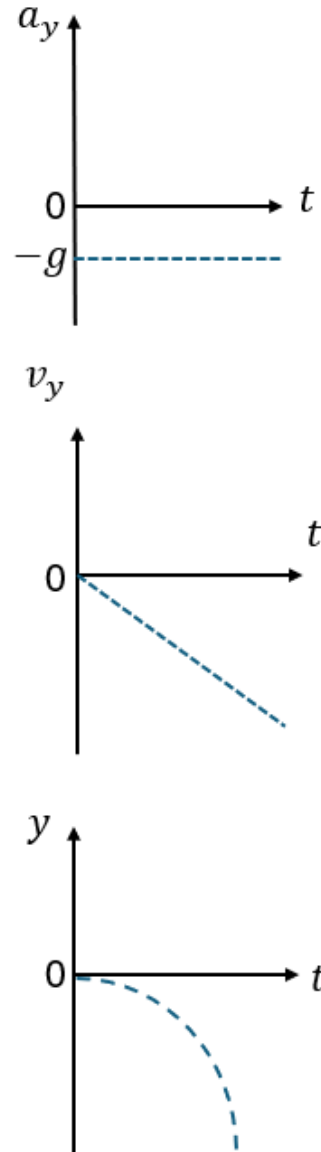


Free Fall

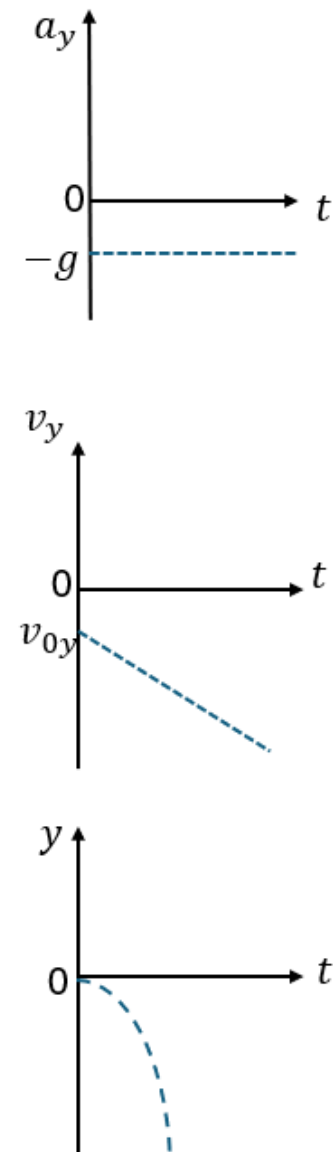
Initial velocity upward



Initial velocity zero



Initial velocity downward



* Location where initial velocity is applied is the origin.

Example 5: Dropping a ball

A physics student dropped a ball from rest from a height of 13.5 m. How long does it take the brick to reach the ground? Neglect air resistance.

Answer: $t = 1.66 \text{ s}$

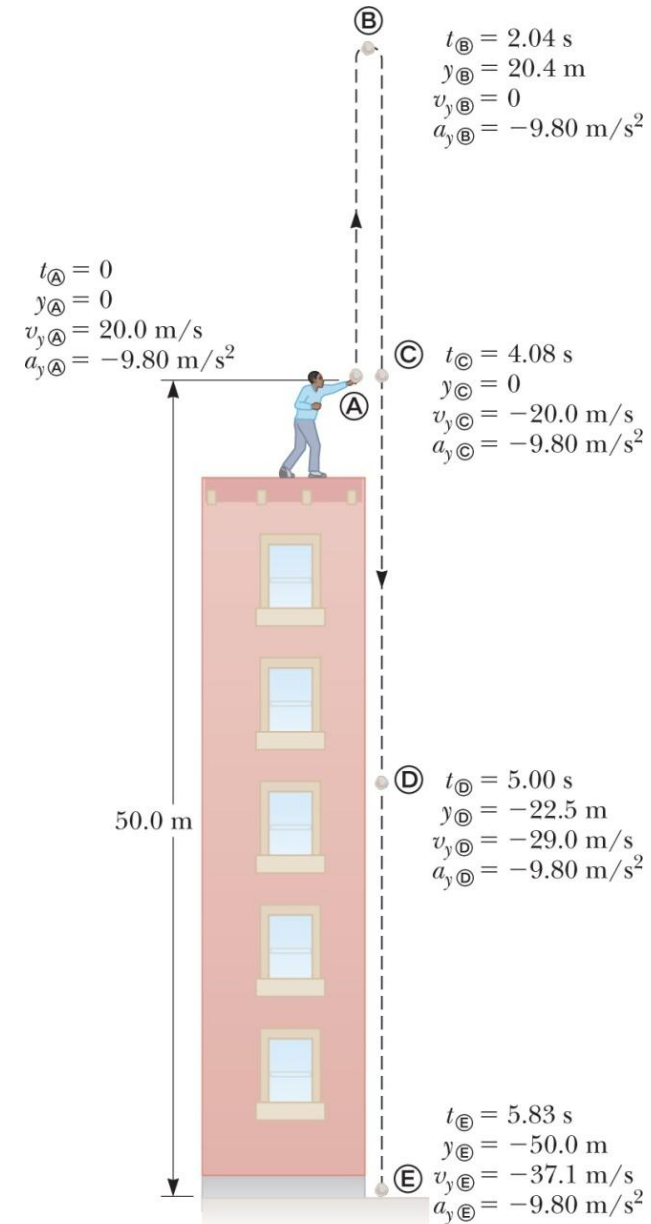
Example 5: Dropping a ball

A ball dropped from rest from a tower 13.5 meters above the ground. Neglecting air resistance, what is the velocity of the ball when it hits the ground? What is its speed?

Answer: $\vec{v} = -16.3 \text{ m/s}$ &
 $|\vec{v}| = 16.3 \text{ m/s}$

Example 6: Stone Throw

A stone thrown from the top of a building is given an initial velocity of 20.0 m/s straight upward. The stone is launched 50.0 m above the ground, and the stone just misses the edge of the roof on its way down as shown in the figure.



Example 6: Stone Throw

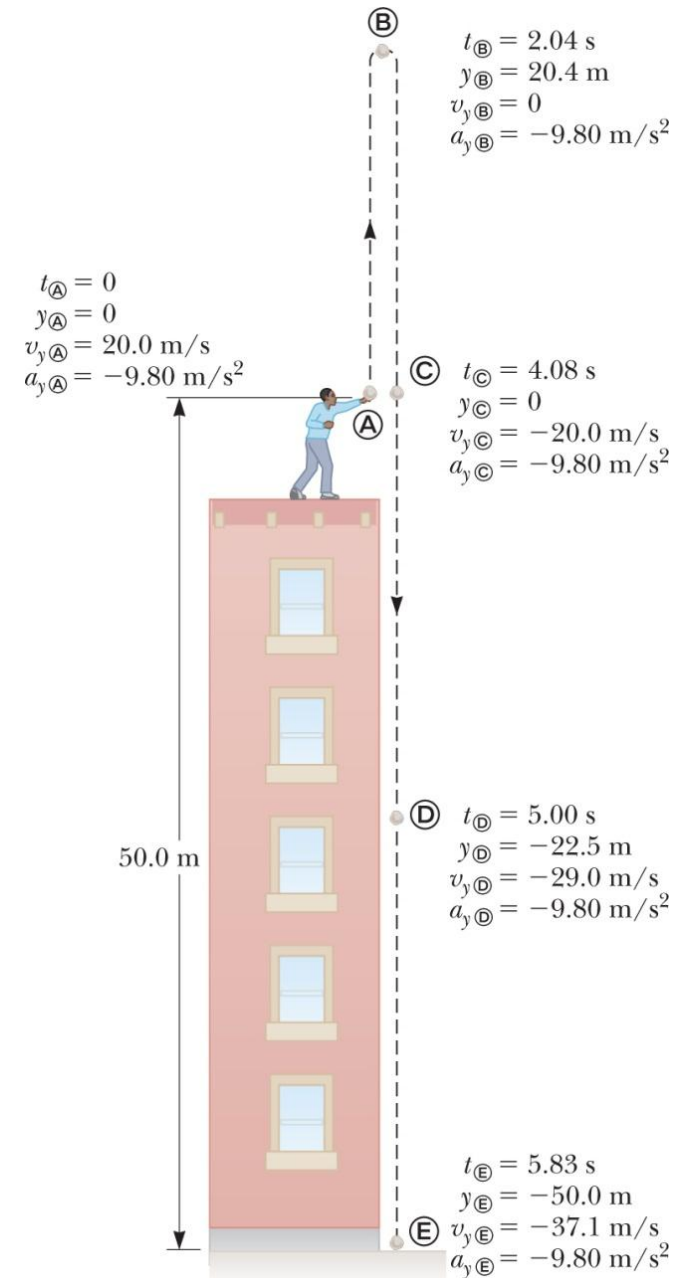
(A) Using $t_A = 0$ as the time the stone leaves the thrower's hand at position A , determine the time at which the stone reaches its maximum height.

$$v = v_0 + a_y t$$

$$\Rightarrow t = \frac{v - v_0}{a_y} = \frac{v_{yB} - v_{yA}}{-g}$$

$$t = t_B = \frac{0 - 20.0 \text{ m/s}}{-9.80 \text{ m/s}^2}$$

$$t_B = \boxed{2.04 \text{ s}}$$



Example 6: Stone Throw

(B) Find the maximum height of the stone (above its initial position).

$$y = y_0 + v_0 t + \frac{1}{2} a_y t^2$$

$$y_{\max} = y_B = y_A + v_A t + \frac{1}{2} a_y t^2$$

$$y_B = 0 + (20.0 \text{ m/s})(2.04 \text{ s}) + \frac{1}{2} (-9.8 \text{ m/s}^2)(2.04 \text{ s})^2$$

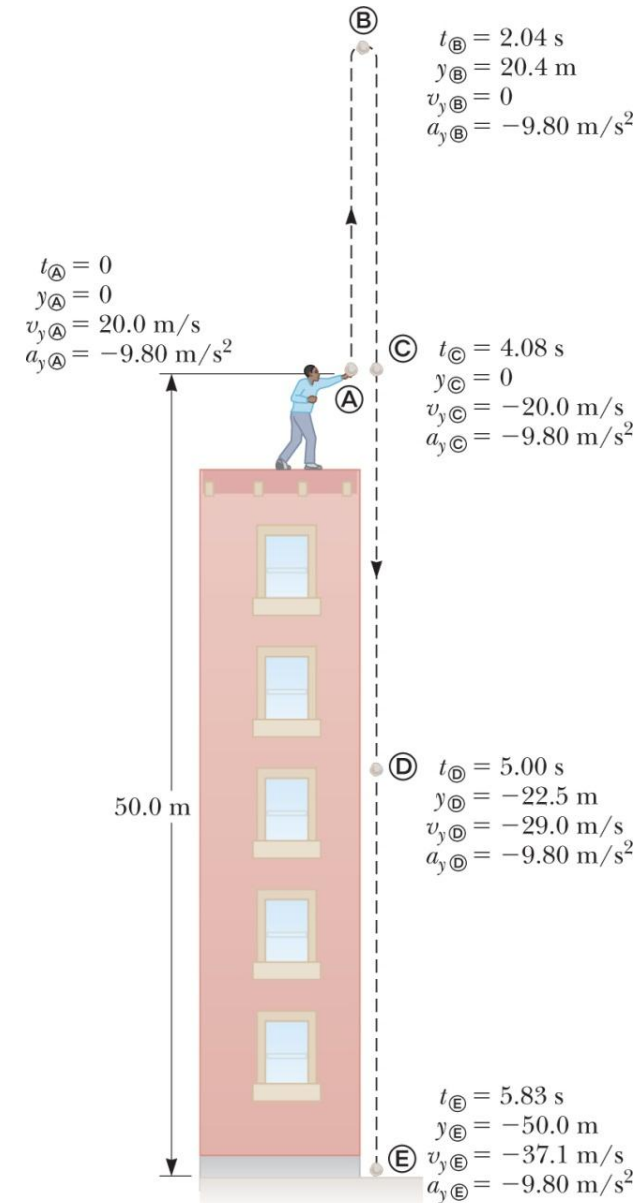
Negative sign!

$$y_B = \boxed{20.4 \text{ m}}$$

You can use,

$$v^2 = v_0^2 + 2a_y(y - y_0)$$

$$v_{yB}^2 = v_{yA}^2 + 2a_y(y_B - y_A)$$



Example 6: Stone Throw

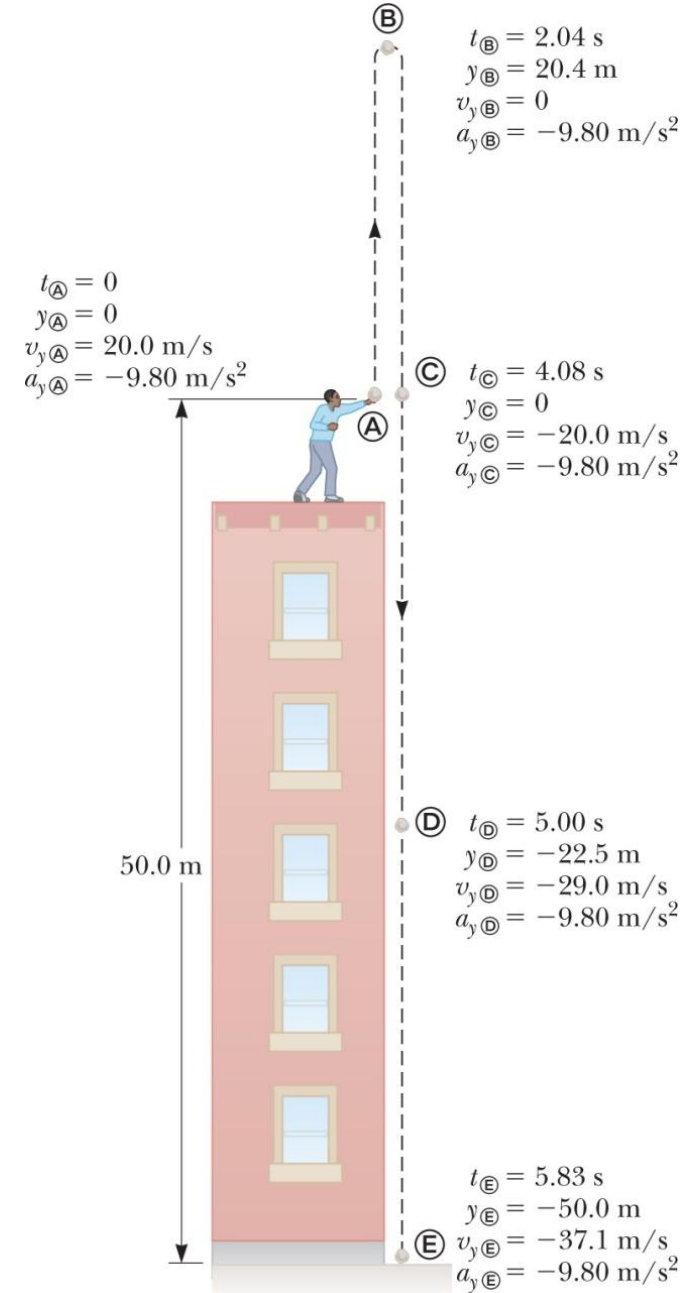
(C) Determine the velocity of the stone when it returns to the height from which it was thrown.

$$v^2 = v_0^2 + 2a_y(y - y_0)$$

$$v_{yC}^2 = v_{yA}^2 + 2a_y(y_C - y_A)$$

$$\begin{aligned} v_{yC}^2 &= (20.0 \text{ m/s})^2 + 2(-9.80 \text{ m/s}^2)(0 - 0) \\ &= 400 \text{ m}^2/\text{s}^2 \end{aligned}$$

$$v_{yC} = \boxed{-20.0 \text{ m/s}} \quad \text{Why negative root?}$$



Example 6: Stone Throw

(D) Find the velocity and position of the stone at $t = 5.00$ s

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a}_y t$$

$$v_{yD} = v_{yA} + a_y t$$

$$= 20.0 \text{ m/s} + (-9.80 \text{ m/s}^2)(5.00 \text{ s})$$

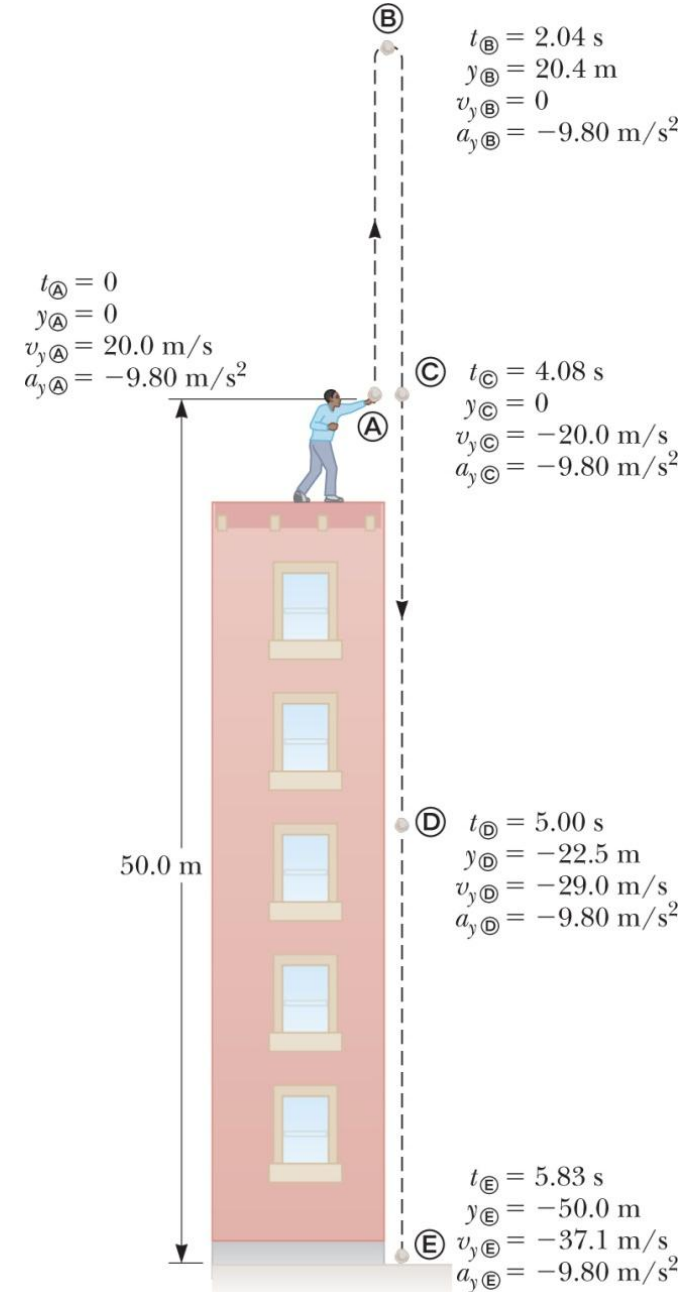
$$= \boxed{-29.0 \text{ m/s}}$$

$$y = y_0 + v_0 t + \frac{1}{2} a_y t^2$$

$$y_D = y_A + v_{yA} t + \frac{1}{2} a_y t^2$$

$$= 0 + (20.0 \text{ m/s})(5.00 \text{ s}) + \frac{1}{2} (-9.80 \text{ m/s}^2)(5.00 \text{ s})^2$$

$$= \boxed{-22.5 \text{ m}}$$



Chapter 3 Summary: Concepts

Displacement

$$\Delta x = x_f - x_i$$

Total displacement

$$\Delta x_{\text{Total}} = \sum \Delta x_i$$

Average velocity (for constant acceleration)

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Instantaneous velocity

$$v(t) = \frac{dx(t)}{dt}$$

Average speed

$$\text{Average speed} = \bar{s} = \frac{\text{Total distance}}{\text{Elapsed time}}$$

Instantaneous speed

$$\text{Instantaneous speed} = |v(t)|$$

Average acceleration

$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}$$

Instantaneous acceleration

$$a(t) = \frac{dv(t)}{dt}$$

Chapter 3 Summary: Formulas

- **1D Motion Kinematics:**

Position from average velocity

$$x = x_0 + \bar{v}t$$

Average velocity

$$\bar{v} = \frac{v_0 + v}{2}$$

Velocity from acceleration

$$v = v_0 + at \quad (\text{constant } a)$$

Position from velocity and acceleration

$$x = x_0 + v_0t + \frac{1}{2}at^2 \quad (\text{constant } a)$$

Velocity from distance

$$v^2 = v_0^2 + 2a(x - x_0) \quad (\text{constant } a)$$

- **Free-Fall Motion:** $a_y = -g = -9.80 \text{ m/s}^2$ (downward):

Velocity of free fall

$$v = v_0 - gt \quad (\text{positive upward})$$

$$v = v_y$$

Height of free fall

$$y = y_0 + v_0t - \frac{1}{2}gt^2$$

$$a = a_y = -g$$

Velocity of free fall from height

$$v^2 = v_0^2 - 2g(y - y_0)$$

Find the direction of Acceleration

(1) Graphical method: slope of the tangent line in the velocity–time (v_x – t) graph.

(2) Definition:

$$\bar{a} \equiv \frac{\Delta v}{\Delta t} = \frac{v_f - v_0}{t_f - t_0}$$

➤ Draw v_f and v_0 ; then determine Δv and its direction, which is the direction of acceleration.

(3) In 1-D motions, if speed is **increasing**, v_x and a_x have the **same direction**; if speed is **decreasing**, v_x and a_x have **opposite** directions.